

```

y<-matrix(c(73,50,128,170,87,108,135,69,148,132), ncol = 1)
y
x_mat<-c(30,20,60,80,40,50,60,30,70,60)
x_mat
x<-cbind(1,x_mat)
x
dim(y)
dim(x)
beta_hat<-solve(t(x)%*% x) %*%t(x) %*% y
beta_hatY
y_hat <- x %*% beta_hat
y_hat
e<- y- y_hat
e
SSE <- t(e) %*% e
SSE
qqnorm(e)
model<-lm(y~ x)
y<-c(73,50,128,170,87,108,135,69,148,132)
y
x<-c(30,20,60,80,40,50,60,30,70,60)
x
model<-lm(y~ x)
model
summary(model)
summary.aov(model)
yield<-c(23,26,20,17, 18,28,17,21, 16,25,12,14)
variety<-c('A','A','A','A', 'B','B','B','B', 'C','C','C','C')
variety<- as.factor(variety)
model<- aov(yield~variety)
model
summary(model)
tapply(yield, variety, mean)
TukeyHSD(model)
tapply(yield, variety, mean)
x<-c(1,2,3,4,5,6)
x
y<-c(4,6,8,10,12,16)
y
plot(x,y)
cor(x,y)
scatter.smooth(x,y)
barplot(x,y)
boxplot(x,y)
mean(x)
var(x)
sd(x)
max(x)
min(x)
IQR(min(x),-max(x))
range(max(x),-max(x))
matrix<-c(2,4,6,8,10,12), byrow = FALSE)
IQR(x)
range(max(x)-min(x))
summary(x)
quantile(x,0.5)
quantile(x,0.25)
quantile(x,0.75)
data<-iris
data
head(data)
data
str(data)
min(data$Sepal.Length)

```

```

data<-cars
data
head(data)
str(data)
names(data)
model<-lm(dist~speed,data = cars)
model
plot(scatter.smooth)
scatter.smooth(cars)
A<-matrix(c(2,4,6,8,10,12), ncol = 3, byrow = TRUE)
A
rownames(A)<- c('R1', 'R2','R3')
colnames(A)<- c('C1','C2')
A
S<-matrix(c(3,4,5,6), nrow = 2, byrow = TRUE)
S
det(S)
t(S)
a<-matrix(c(5,4,3,2,6,7), nrow = 3, ncol = 3, byrow = TRUE)
a
b<-matrix(c(7,8,9,5,4,3), nrow = 3, ncol = 3, byrow = FALSE)
b
a+b
a-b
a%%b
a*b
v<- matrix(c(2,4,6,7,8), nrow = 1)
v
t(v)
A<-matrix(c(2,4,5,6,7,5,4,3), nrow = 4, ncol = 4, byrow = TRUE)
A
B<-matrix(c(7,4,2,7,4,6,2,8), nrow = 4, ncol = 4, byrow = FALSE)
B
A%%B
A+B
A-B
det(A)
det(B)
t(A)
t(B)
D<-matrix(c(23,0,0,0,45,0,0,0,67), nrow = 3, byrow = TRUE)
D
I<-matrix(c(1,0,0,0,0,1,0,0,0,0,1,0,0,0,0,1,0,0,0,0), nrow = 8, byrow = TRUE)
I
t(I)
det(I)

```